YEAR 11 PHYSICS, AQA GCSE, LESSON 5 OF 10

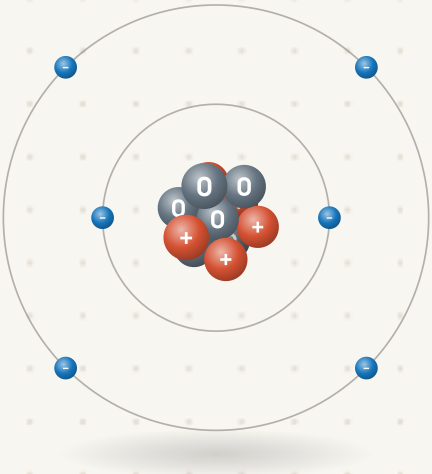
# Atoms & Nuclear Radiation.

## Lesson 05 Radioactive decay

A smoke alarm contains a radioactive source. Nothing switches it on, yet it gives out radiation for hundreds of years. Why?

### IN THIS LESSON YOU WILL

- explain why some nuclei give out radiation
- describe radioactive decay as random
- define activity and count rate
- describe what alpha, beta, gamma and neutron radiation are, and how each changes the nucleus



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

# How this booklet works.

This is the booklet for **Lesson 5** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

**DO NOW**
Quick questions from earlier lessons. Remembering again is what makes it stick.

**LEARN IT**
Explanations, diagrams and worked examples. Read this again when you revise.

**TRY IT**
Practice that builds up: first with support, then on your own, then a challenge.

**LAB**
Record what you find in a simulation or a model experiment.

**EXAM CORNER**
AQA-style questions with a tip on how the marks are given.

**REVISE IT**
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

## The route through the unit (this lesson is highlighted)


01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

**05 Radioactive decay**
Unstable nuclei, activity and count rate, alpha, beta, gamma and neutron radiation (slides 26–30)

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation


Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

# 05 Radioactive decay.

AQA Physics 4.4.2.1; Combined Science 6.4.2.1

BIG QUESTION

**A smoke alarm contains a radioactive source. Nothing switches it on, yet it gives out radiation for hundreds of years. Why?**

BY THE END YOU CAN

- explain why some nuclei give out radiation
- describe radioactive decay as random
- define activity and count rate
- describe what alpha, beta, gamma and neutron radiation are, and how each changes the nucleus

unstable nucleus | radioactive decay | random | activity | becquerel (Bq) | count rate | Geiger–Müller tube | alpha | beta | gamma

SLIDES AND SIMULATIONS

**Teaching deck, slides 26–30**

Radioactive decay, activity and count rate, then alpha, beta and gamma.

panphy.app/gcsephy/decks/atoms-and-radiation/#26

**Nuclear Decay simulation**

panphy.app/simulations/nuclear_decay.html

RETRIEVE

## Do now

From earlier lessons. No notes!

1 What is an isotope?

2 What happens when an electron emits EM radiation?

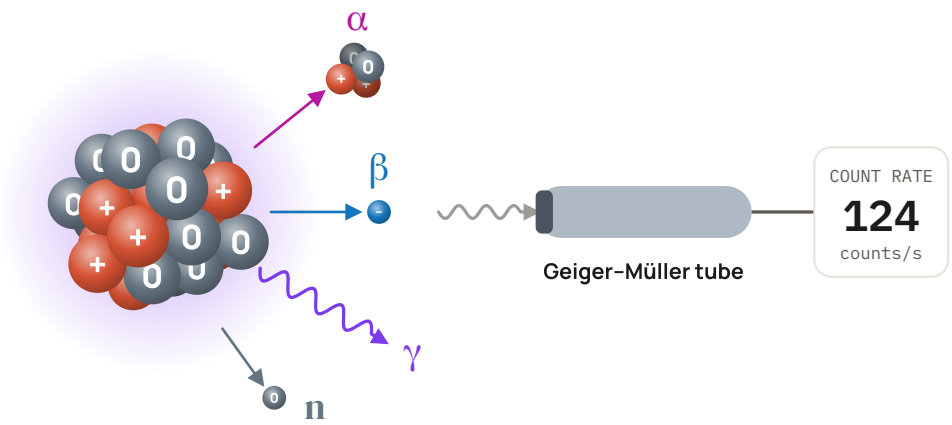
3 How many protons and neutrons are in $^{14}_{6}C$?

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EXPLAIN

# When a nucleus is unstable

## Unstable nuclei




Some isotopes have an **unstable nucleus**. It gives out **radiation** as it changes to become **more stable**. This is **radioactive decay**.

Decay is **random**: you cannot predict which nucleus will decay next, or when. Nothing we do changes it: heating, cooling or chemical reactions have no effect.

| Quantity | Meaning | Unit |
|---|---|---|
| **Activity** | the rate at which a source of unstable nuclei decays: the number of decays each second | becquerel (Bq); 1 Bq = 1 decay per second |
| **Count rate** | the number of decays recorded each second by a detector, such as a Geiger–Müller tube | counts per second (or per minute) |

THINK OF POPCORN

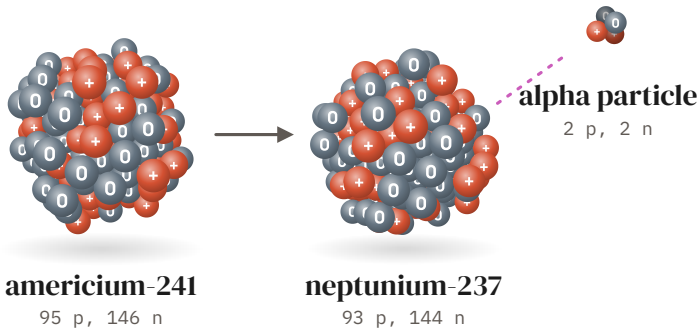
In a hot pan, you cannot tell which kernel will pop next, or when. But with lots of kernels, you can predict roughly how many will pop each minute.

**Where the model breaks:** heating makes popcorn pop faster, but nothing we do changes the rate of radioactive decay.

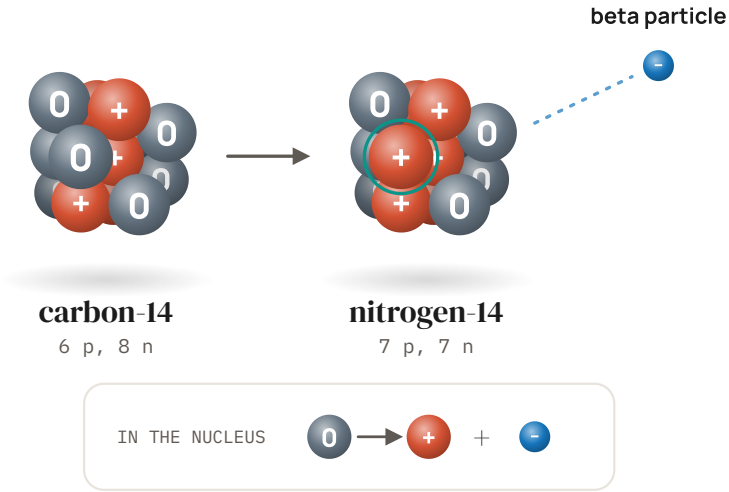
## Four kinds of nuclear radiation

| Radiation | What it is | Symbol | Charge | What happens in the nucleus |
|---|---|---|---|---|
| **Alpha (α)** | 2 protons + 2 neutrons: a helium nucleus | $^{4}_{2}He$ | +2 | loses 2 protons and 2 neutrons |
| **Beta (β)** | a high-speed electron ejected from the nucleus | $^{0}_{-1}e$ | −1 | a neutron turns into a proton |
| **Gamma (γ)** | electromagnetic radiation from the nucleus | $\gamma$ | 0 | loses energy only |
| **Neutron (n)** | a neutron | $^{1}_{0}n$ | 0 | loses one neutron |

## Alpha decay

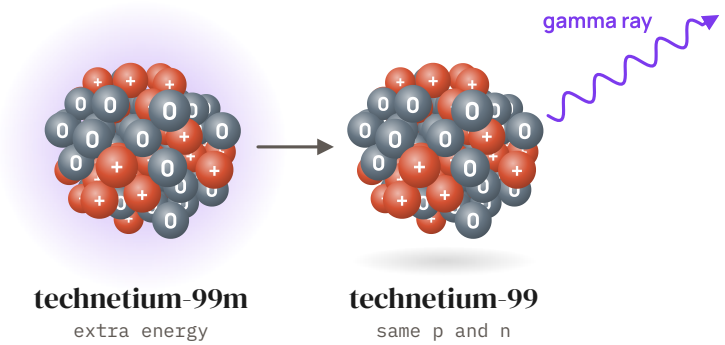



## Beta decay




In beta decay, a **neutron changes into a proton**, and a fast electron is made and shoots out of the nucleus. The beta particle comes from the **nucleus**, not from the energy levels.

## Gamma emission




Gamma rays are **electromagnetic waves**. The nucleus gives out extra energy, but its protons and neutrons do not change. Gamma often follows alpha or beta decay.

PRACTISE

### Try it

**1** Draw a line from each type of radiation to what it is.

| | |
|---|---|
| **1** alpha (α) | **A** a neutron |
| **2** beta (β) | **B** a high-speed electron from the nucleus |
| **3** gamma (γ) | **C** two protons and two neutrons |
| **4** neutron (n) | **D** electromagnetic radiation |

**2** Which statement about radioactive decay is correct? Tick **one** box.

- ☐ Heating a source makes it decay faster.
- ☐ You can predict when a particular nucleus will decay.
- ☐ Unstable nuclei decay at random to become more stable.
- ☐ A nucleus decays when a detector is placed near it.

**3** Complete the sentences.

The _____ of a source is the number of decays each second. It is measured in _____ (Bq). The number of decays a Geiger–Müller tube records each second is the _____.

WORD BANK: **activity** | **count rate** | **becquerels** | **half-life** | **seconds**

**4** SUPPORTED Which radiation causes each change in the nucleus?

| Change in the nucleus | Radiation |
|---|---|
| loses 2 protons and 2 neutrons | |
| a neutron turns into a proton | |
| loses energy, but no particles change | |
| loses one neutron | |

**5** Explain what is wrong with Ava's statement.

**Ava:** Beta particles are electrons knocked out of the energy levels, so beta decay just turns the atom into an ion.

**6** CHALLENGE A Geiger–Müller tube records 1800 counts in 1 minute. (a) Calculate the count rate in counts per second. (b) Suggest why the activity of the source is greater than this count rate.

INVESTIGATE

## Virtual lab: watch a nucleus decay

Nuclear Decay simulation, tabs 01 to 03

Open the **Nuclear Decay** simulation. For each tab, press **Decay this nucleus** and read the equation it shows.

**7** Record what happens.

| Tab | Parent nucleus | What leaves the nucleus? | Change in mass number | Change in atomic number | New nucleus |
|---|---|---|---|---|---|
| 01 α | americium-241 | | | | |
| 02 β | carbon-14 | | | | |
| 03 γ | technetium-99m | | | | |

**8** Which decays make a new element? Explain using your table.

AQA-STYLE

## Exam corner

Write in full sentences. Show every step of a calculation.

**Get the marks:** Two marks: say what the nucleus gives out *and* why.

**9** Describe what is meant by radioactive decay. **[2 marks]**

**10** Compare an alpha particle with a beta particle. **[3 marks]**

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REVIEW LATER

# Revise it

Use this box after the lesson and before tests.

## SUMMARY

- Unstable nuclei give out radiation to become more stable: radioactive decay.
- Decay is random: we cannot predict which nucleus decays, or when.
- Activity: decays per second, in becquerel (Bq). Count rate: decays recorded per second by a detector.
- Alpha: 2p + 2n (helium nucleus). Beta: fast electron; a neutron becomes a proton. Gamma: EM radiation. Neutron: a neutron.

## COVER THE PAGE AND ANSWER

**1** Why do some nuclei give out radiation?

**2** What is the unit of activity?

**3** What is an alpha particle?

**4** What happens in the nucleus in beta decay?

## I CAN…

- explain why unstable nuclei decay
- describe decay as random
- define activity and count rate
- describe alpha, beta, gamma and neutron radiation

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REFERENCE

# Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

## The three particles

Lesson 1

| Particle | Charge | Mass | Where |
|---|---|---|---|
| **Proton** | +1 | 1 | nucleus |
| **Neutron** | 0 | 1 | nucleus |
| **Electron** | −1 | very small | energy levels |

## Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ $1 \times 10^{-10}$ m; nucleus radius < 1/10 000 of that

## Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

## Alpha scattering

Lesson 3

| Observation | Conclusion |
|---|---|
| **most straight through** | mostly empty space |
| **some deflected** | positive centre |
| **a very few bounced back** | mass in a tiny centre |

## Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

## Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

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REFERENCE

# Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

**Activity**: The rate at which a source of unstable nuclei decays. Unit: becquerel (Bq).

**Alpha particle (α)**: Two protons and two neutrons: a helium nucleus. Strongly ionising; stopped by paper.

**Becquerel (Bq)**: Unit of activity: 1 Bq = 1 decay per second.

**Beta particle (β)**: A high-speed electron ejected from the nucleus when a neutron turns into a proton.

**Count rate**: The number of decays recorded each second by a detector.

**Gamma ray (γ)**: Electromagnetic radiation emitted from a nucleus. Weakly ionising; very penetrating.

**Radioactive decay**: An unstable nucleus giving out radiation to become more stable. It is random.

**Unstable nucleus**: A nucleus that will decay and give out radiation.

CHECK YOURSELF

# Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

### DO NOW

1. An atom of the same element with a different number of neutrons.
2. It moves to a lower energy level, closer to the nucleus.
3. 6 protons, 8 neutrons.

### COVER THE PAGE

1. They are unstable; they decay to become more stable.
2. Becquerel (Bq).
3. 2 protons and 2 neutrons (a helium nucleus).
4. A neutron turns into a proton (and a fast electron is emitted).

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